Math 110 — Week 1

Proofs, sets, fields, vector spaces, subspaces (Notes Ch. 1–2, LADR 1A–1C, HW1)

§1 This week

§1.1 How proofs are written (notes, opening section)

The question. A proof is a list of statements in which each one is an assumption, a definition unfolded, or follows from earlier lines by a rule of logic. Which standard shapes do such lists take?

Direct proof: unfold the definition, compute, recognise the definition again. Claim: the product of two odd integers is odd.

1. "Odd" means of the form $2k+1$. Write $m = 2a+1$, $n = 2b+1$.
2. $mn = (2a+1)(2b+1) = 2(2ab+a+b)+1$.
3. $2ab+a+b$ is an integer, so mn has the form $2k+1$: it is odd.

The definition is used twice: once to turn the hypothesis into something to calculate with, once to know what the answer has to look like. With $m = n$: m odd $\Rightarrow m^2$ odd.

Contrapositive. $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are true or false together. The contrapositive of "m odd $\Rightarrow m^2$ odd" is "m^2 not odd $\Rightarrow m$ not odd", i.e. m^2 even $\Rightarrow m$ even, with no new work.

If and only if means two implications, and each may need its own argument. Example: m even $\iff m^2$ even. ($\Leftarrow$) is the contrapositive above. ($\Rightarrow$): $m = 2k$ gives $m^2 = 2(2k^2)$.

Proposition (Prop. 1)
$\sqrt{2}$ is irrational.

The idea

"Irrational" is a negative statement (there is *no* fraction equal to $\sqrt{2}$), so there is nothing to compute with directly. Assuming the opposite gives an equation: $\sqrt{2} = m/n$, so $m^2 = 2n^2$. The useful property of a fraction is that it can be written in lowest terms, so m, n are not both even. The equation $m^2 = 2n^2$ talks about evenness, and "m^2 even $\Rightarrow m$ even" is available. So follow evenness through the equation and aim to show both m and n are even.

Proof. 1. Suppose $\sqrt{2} = m/n$ with integers m, n, $n \neq 0$, in lowest terms. So at most one of m, n is even.
2. Squaring: $m^2 = 2n^2$. So m^2 is even, hence m is even. Write $m = 2r$.
3. Then $4r^2 = 2n^2$, so $n^2 = 2r^2$ is even, hence n is even.
4. Both m and n are even, contradicting step 1.

□

Equality of sets: two inclusions. $A = B$ means the same elements, so prove $A \subseteq B$ and $B \subseteq A$, each by taking an arbitrary element. Example: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

1. ($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$, and $x \in B$ or $x \in C$. In the first case $x \in A \cap B$, in the second $x \in A \cap C$.
2. ($\supseteq$) Let $x \in (A \cap B) \cup (A \cap C)$. In either case $x \in A$, and $x \in B$ or $x \in C$, so $x \in A \cap (B \cup C)$.

Proposition (Prop. 2: induction)
Let $P(n)$ be a statement for each integer $n \geq 1$. If $P(1)$ holds, and $P(n) \Rightarrow P(n+1)$ for every $n \geq 1$, then $P(n)$ holds for all $n \geq 1$.

Use it when the case $n+1$ can be reduced to the case n. In this course that is usually induction on a dimension or on the length of a list.

§1.2 Sets and functions (Ch. 1)

Definition (Def. 1, 2)**.** A **set** is a collection of distinguishable objects, its **elements**. $x \in M$: x is an element of M. $N \subseteq M$: every element of N is in M. A **function** $f\colon M \to N$ assigns an element $f(x) \in N$ to every $x \in M$.

- Ex. 1: $\{x : x \in \{1, \dots, 6\},\ x \text{ even}\} = \{2, 4, 6\}$.
- For subsets M, N of A: complement $M^c = \{x \in A : x \notin M\}$; union $M \cup N$ ($x \in M$ or $x \in N$); intersection $M \cap N$ ($x \in M$ and $x \in N$); difference $M \setminus N = M \cap N^c$; product $M \times N = \{(x, y) : x \in M, y \in N\}$. Pairs are ordered: $(1, 2) \neq (2, 1)$.
- f is **injective** if $f(x) = f(y) \Rightarrow x = y$; **surjective** if every $z \in N$ is $f(x)$ for at least one x; **bijective** if both.

Proposition (Ex. 2: pigeonhole principle)
If A, B are finite sets with the same number of elements and $f\colon A \to B$, then f is injective $\iff$ f is surjective.

The idea

Count the values f takes. Say A and B have n elements. f produces n values $f(a)$, one for each a. If two of them coincide, at most $n-1$ different elements of B are reached, so one is missed. If none coincide, n different elements of B are reached, which is all of B.

This fails for infinite sets: $f(n) = n+1$ on $\{1, 2, 3, \dots\}$ is injective but misses 1. Later, for linear maps between spaces of the same finite dimension, the same equivalence holds, with dimension in the role of the number of elements.

Proof. 1. Let n be the number of elements of A and of B, and let k be the number of different values among the $f(a)$. So $k \leq n$.
2. f is injective exactly when no two values coincide, i.e. $k = n$.
3. f is surjective exactly when all n elements of B are values, i.e. $k = n$.

□

§1.3 Fields (Ch. 1)

The question. Which number systems are allowed as scalars? We need to add, subtract, multiply, and divide by anything nonzero. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ allow this; $\mathbb{Z}$ does not (2 has no inverse in $\mathbb{Z}$).

Definition (Def. 3)**.** A **field** is a set $\mathbb{F}$ with two different elements $0, 1$ and operations $+, \cdot$ such that for all $x, y, z \in \mathbb{F}$:

(i) $x + y = y + x$ and $xy = yx$;
(ii) $x + (y + z) = (x + y) + z$ and $x(yz) = (xy)z$;
(iii) $x + 0 = x$;
(iv) $1 \cdot x = x$;
(v) $x(y + z) = xy + xz$;

(vi) every x has an additive inverse $-x$ with $(-x)+x=0$;
(vii) every $x\neq 0$ has a multiplicative inverse x^{-1} with $x^{-1}x=1$.

Ex. 3: $\mathbb{Q}$ and $\mathbb{R}$ are fields. $\mathbb{Z}$ is not: only ± 1 have multiplicative inverses in $\mathbb{Z}$.

Proposition (Uniqueness of identities and inverses)
In a field, the additive identity, the multiplicative identity, additive inverses and multiplicative inverses are unique.

The idea

To show there is only one object with a property, take two objects with it and show they are equal. For identities: combine the two candidates, $0+0'$. Read with $0'$ as the identity, this is 0; read with 0 as the identity, this is $0'$. For inverses: put x between its two inverses, $(-x)+x+(-x)'$, and group it in two ways.

Proof. 1. Let $0'$ also satisfy $x+0'=x$ for all x. Then $0=0+0'$ (property of $0'$) $=0'+0$ (axiom (i)) $=0'$ (axiom (iii)).
2. The same argument with multiplication gives $1=1\cdot 1'=1'$.
3. If $-x$ and $(-x)'$ are both additive inverses of x: $-x=-x+0=-x+(x+(-x)')=(-x+x)+(-x)'=0+(-x)'=(-x)'$.
4. The same argument with multiplication gives uniqueness of x^{-1}.

□

Proposition (HW1 P3: no zero divisors)
In a field, $0\cdot a=0$ for all a, and if $ab=0$ then $a=0$ or $b=0$.

Proof. 1. $0\cdot a=(0+0)a=0\cdot a+0\cdot a$. Add $-(0\cdot a)$ to both sides: $0=0\cdot a$.
2. Suppose $ab=0$ and $a\neq 0$. Multiply by a^{-1}: $b=1\cdot b=(a^{-1}a)b=a^{-1}(ab)=a^{-1}\cdot 0=0$.

□

The second step uses axiom (vii). In $\mathbb{Z}/6$ (not a field) $2\cdot 3=0$.

Complex numbers (Ex. 4). $\mathbb{C}=\{a+bi : a,b\in\mathbb{R}\}$ with $i^2=-1$; add and multiply by the usual rules. $\mathbb{R}\subseteq\mathbb{C}$ via $x=x+0i$.

- $(1+i)^3=1+3i+3i^2+i^3=1+3i-3-i=-2+2i$. $\quad(5+i)(4+3i)=20+15i+4i+3i^2=17+19i$.
- **Conjugate**: $\overline{a+bi}=a-bi$. Then $(a+bi)(a-bi)=a^2+b^2$.
- **$\mathbb{C}$ is a field.** The other axioms follow from the rules of arithmetic; inverses need a formula. If $a+bi\neq 0$ then $a^2+b^2>0$, and $(a+bi)\cdot\dfrac{a-bi}{a^2+b^2}=\dfrac{a^2+b^2}{a^2+b^2}=1$. Example: $\dfrac{1}{3+4i}=\dfrac{3-4i}{25}$.
- **Why $\mathbb{C}$ matters.** The cubic $x^3-15x-4=0$ has the real root 4, but the cubic formula gives $\sqrt[3]{2+11i}+\sqrt[3]{2-11i}$. Since $(2\pm i)^3=2\pm 11i$, this is $(2+i)+(2-i)=4$: complex numbers appear inside a calculation whose answer is real. Later: every nonconstant complex polynomial has a complex root (fundamental theorem of algebra), which is why operators on complex spaces always have eigenvalues and operators on real spaces need not.

Proposition (Prop. 3: conjugation)
For $w,z\in\mathbb{C}$: (i) $\overline{w+z}=\bar{w}+\bar{z}$; (ii) $\overline{wz}=\bar{w}\,\bar{z}$; (iii) $\bar{z}=z \iff z$ is real; (iv) $z\bar{z}$ is real and ≥ 0.

Proof. Write $w=a+bi$, $z=c+di$.

1. $\overline{w+z}=(a+c)-(b+d)i=(a-bi)+(c-di)$.
2. $wz=(ac-bd)+(ad+bc)i$, and $\bar{w}\bar{z}=(a-bi)(c-di)=(ac-bd)-(ad+bc)i$.

3. $z - \bar{z} = 2di$, which is 0 exactly when $d = 0$.
4. $z\bar{z} = c^2 + d^2 \geq 0$.

□

The field $\mathbb{F}_2$. $\{0, 1\}$ with $1 + 1 = 0$ (and the other sums and products as usual) is a field. In it $2x = x + x = 0$ for every x, so $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$ (HW1 P7). Most theorems of linear algebra hold over any field; the difference between $\mathbb{R}$ and $\mathbb{C}$ matters once eigenvalues appear.

§1.4 Vector spaces (Ch. 2)

The question. Tuples, functions, sequences and polynomials can all be added and multiplied by numbers. Which rules about these two operations are enough to prove theorems for all of them at once?

Definition (Def. 4 / LADR 1.20)**.** An **$\mathbb{F}$-vector space** is a set V with addition $V \times V \to V$ and scalar multiplication $\mathbb{F} \times V \to V$ such that for all $u, v, w \in V$ and $\lambda, \mu \in \mathbb{F}$:

(i) $v + w = w + v$;
(ii) $(v + w) + u = v + (w + u)$;
(iii) there is $0 \in V$ with $v + 0 = v$ for all v;
(iv) every v has $-v \in V$ with $v + (-v) = 0$;
(v) $(\lambda\mu)v = \lambda(\mu v)$;
(vi) $\lambda(v + w) = \lambda v + \lambda w$ and $(\lambda + \mu)v = \lambda v + \mu v$;
(vii) $1 \cdot v = v$.

- The zero vector and additive inverses are unique, by the same proofs as for a field (LADR 1.26, 1.27).
- Axiom (vii) is needed: without it, "$\lambda v = 0$ for all λ, v" would satisfy (i)–(vi).
- Ex. 5: $\mathbb{F}^n$, lists $(x_1, \ldots, x_n)$ with componentwise operations. Zero vector $(0, \ldots, 0)$; do not confuse it with the number 0. $\mathbb{F} = \mathbb{F}^1$.
- Ex. 6: $\mathbb{F}^S$, all functions $f\colon S \to \mathbb{F}$, with $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$. Zero vector: the constant function 0. For $S = \{1, \ldots, n\}$ this is $\mathbb{F}^n$; for $S = \mathbb{N}$ it is the space of sequences $\mathbb{F}^\infty$.
- Ex. 7: $\mathbb{C}$ is a real vector space (allow only real scalars). $\mathbb{C}$ and $\mathbb{R}$ are $\mathbb{Q}$-vector spaces.

Proposition (Prop. 4 / LADR 1.30–1.32)
(i) $0 \cdot v = 0$ for every $v \in V$. (ii) $\lambda \cdot 0 = 0$ for every $\lambda \in \mathbb{F}$. (iii) $(-1)v = -v$.

The idea

These are not axioms, so they need proofs. The only axiom that connects the number 0 (or addition of numbers) with vectors is the distributive law (vi). So write 0 as $0 + 0$ and distribute: this produces an equation of the form $x = x + x$, and subtracting x leaves $x = 0$. For (iii), $-v$ is *defined* as the vector that gives 0 when added to v, so check that $(-1)v$ does that.

Proof. 1. $0 \cdot v = (0 + 0)v = 0 \cdot v + 0 \cdot v$. Add $-(0 \cdot v)$ to both sides: $0 = 0 \cdot v$.
2. $\lambda \cdot 0 = \lambda(0 + 0) = \lambda \cdot 0 + \lambda \cdot 0$. Add $-(\lambda \cdot 0)$: $0 = \lambda \cdot 0$.
3. $(-1)v + v = (-1)v + 1 \cdot v = (-1 + 1)v = 0 \cdot v = 0$ by (i). So $(-1)v$ is the additive inverse of v, which is unique.

□

Proposition (HW1 P4 / LADR 1B.2)
If $av = 0$ with $a \in \mathbb{F}$, $v \in V$, then $a = 0$ or $v = 0$.

Proof. If $a \neq 0$: $v = 1 \cdot v = (a^{-1}a)v = a^{-1}(av) = a^{-1} \cdot 0 = 0$ (Prop. 4(ii)). □

§1.5 Subspaces (Ch. 2)

The question. Which subsets of a vector space are vector spaces themselves, with the same operations? In $\mathbb{R}^2$: the line $y = 2x$ is; the line $y = 2x + 1$ is not (it misses 0); the union of the two axes is not ($(1,0) + (0,1)$ leaves it).

Definition (Def. 5 / LADR 1.34)**.** A subset $U \subseteq V$ is a **subspace** if (i) $0 \in U$, and (ii) $\lambda v + w \in U$ for all $v, w \in U$, $\lambda \in \mathbb{F}$.

- Condition (ii) is "closed under addition" ($\lambda = 1$) and "closed under scalar multiplication" ($w = 0$) together. A subspace is a vector space: the axioms hold in U because they hold in V, and $-v = (-1)v \in U$.
- **To show a set is a subspace**: check 0, then take v, w in the set and check $\lambda v + w$. **To show it is not**: exhibit $0 \notin U$, or two specific vectors whose sum leaves U, or a specific multiple that leaves U.
- Ex. 8: in $\mathbb{R}^{(0,1)}$, the continuous functions form a subspace, and so do the differentiable functions (sums and multiples of continuous functions are continuous).
- Ex. 9: sequences $(a_0, a_1, \dots)$ with $a_{n+2} = a_{n+1} + a_n$ for all n form a subspace of $\mathbb{R}^{\mathbb{N}_0}$: if a and b satisfy the recurrence, so does $\lambda a + b$, since $\lambda a_{n+2} + b_{n+2} = (\lambda a_{n+1} + b_{n+1}) + (\lambda a_n + b_n)$. The Fibonacci sequence is one element. Each such sequence is determined by a_0, a_1.

Proposition (Prop. 5 / LADR 1C.10)
If U, W are subspaces of V, so is $U \cap W$.

Proof. $0 \in U$ and $0 \in W$. If $v, w \in U \cap W$, then $\lambda v + w \in U$ and $\lambda v + w \in W$. □

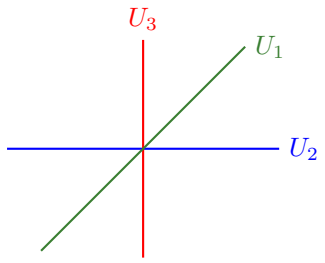
Proposition (Prop. 6 / LADR 1.40)
If U, W are subspaces of V, then $U + W = \{u + w : u \in U, w \in W\}$ is a subspace, and it is the smallest subspace containing $U \cup W$.

The idea

The union $U \cup W$ is usually not a subspace: for the two axes in $\mathbb{R}^2$, $(1,0) + (0,1)$ is in neither. Any subspace containing U and W has to contain every $u + w$. So the set of all such sums is forced; the proposition says that this set is already closed, so nothing more has to be added.

Proof. 1. $0 = 0 + 0 \in U + W$.
2. For $u_1 + w_1$, $u_2 + w_2 \in U + W$: $\lambda(u_1 + w_1) + (u_2 + w_2) = (\lambda u_1 + u_2) + (\lambda w_1 + w_2) \in U + W$.
3. It contains $U \cup W$: $u = u + 0$ and $w = 0 + w$.
4. Smallest: if a subspace X contains $U \cup W$, then for $u \in U$, $w \in W$ both lie in X, so $u + w \in X$. Hence $U + W \subseteq X$.

□



Three different lines through 0 in $\mathbb{R}^2$. $U_2 + U_3 = \mathbb{R}^2$, so $U_1 \cap (U_2 + U_3) = U_1$. But $U_1 \cap U_2 = U_1 \cap U_3 = \{0\}$, so $(U_1 \cap U_2) + (U_1 \cap U_3) = \{0\}$. The distributive law for sets fails for subspaces when union is replaced by sum.

Definition (Def. 6)**.** The sum of U and W is **direct**, written $U \oplus W$, if $U \cap W = \{0\}$. A sum $U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_k \in U_k\}$ is direct, written $U_1 \oplus \cdots \oplus U_m$, if $U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}$ for $k = 2, \ldots, m$.

Theorem (Prop. 7 / LADR 1.41, 1.45)

For subspaces $U_1, \ldots, U_m$ of V the following are equivalent:

(i) the sum $U_1 + \cdots + U_m$ is direct;
(ii) every $v \in U_1 + \cdots + U_m$ is $u_1 + \cdots + u_m$ with $u_k \in U_k$ in exactly one way;
(iii) the only way to write $0 = u_1 + \cdots + u_m$ with $u_k \in U_k$ is with every $u_k = 0$.

The idea

Take two subspaces first. If $u + w = u' + w'$, then $u - u' = w' - w$. The left side is in U, the right side in W, so this vector is in $U \cap W$. If $U \cap W = \{0\}$ it is 0, and $u = u'$, $w = w'$. So "intersection 0" is exactly what makes the pieces of a vector unique. Conversely a nonzero $x \in U \cap W$ gives two ways to write 0: $0 + 0$ and $x + (-x)$.

For m subspaces do the same one subspace at a time, starting from the last: move everything except the U_m-parts to one side to get a vector in $U_m \cap (U_1 + \cdots + U_{m-1})$.

For $m \geq 3$, pairwise intersections being $\{0\}$ is *not* enough: the three lines in the figure meet pairwise in $\{0\}$, but $(1,1) = (1,0) + (0,1)$ lies in $U_1 \cap (U_2 + U_3)$, and $0 = (1,1) - (1,0) - (0,1)$ is a nontrivial way to write 0.

Proof. 1. (i)⇒(ii). A decomposition exists by the definition of the sum. Suppose $v = u_1 + \cdots + u_m = \tilde{u}_1 + \cdots + \tilde{u}_m$. Then $u_m - \tilde{u}_m = (\tilde{u}_1 - u_1) + \cdots + (\tilde{u}_{m-1} - u_{m-1})$. The left side is in U_m and the right side in $U_1 + \cdots + U_{m-1}$, so both are in $U_m \cap (U_1 + \cdots + U_{m-1}) = \{0\}$. So $u_m = \tilde{u}_m$.

2. Now $u_1 + \cdots + u_{m-1} = \tilde{u}_1 + \cdots + \tilde{u}_{m-1}$, and the same argument gives $u_{m-1} = \tilde{u}_{m-1}$. Continue down to $u_1 = \tilde{u}_1$.

3. (ii)⇒(iii). $0 = 0 + \cdots + 0$ is one way to write 0; by (ii) it is the only one.

4. (iii)⇒(i), by contrapositive. If the sum is not direct, some $U_k \cap (U_1 + \cdots + U_{k-1})$ contains $u_k \neq 0$, say $u_k = u_1 + \cdots + u_{k-1}$. Then $u_1 + \cdots + u_{k-1} + (-u_k) + 0 + \cdots + 0 = 0$ with $-u_k \neq 0$, contradicting (iii).

□

Shortcuts

- **Subspace test in one line**: $0 \in U$ and $\lambda v + w \in U$. For a set defined by equations, it is a subspace exactly when the equations are linear with right side 0 (e.g. $x_1 + 2x_2 = 0$ yes; $x_1 + 2x_2 = 1$ no; $x_1 x_2 = 0$ no).
- **Fastest disproof**: check 0 first. If $0 \notin U$, stop.
- **Direct sum of two subspaces**: only $U \cap W = \{0\}$ needs checking. For three or more, check that 0 has only the trivial decomposition (Prop. 7(iii)); pairwise intersections are not enough.
- **Dividing by a complex number**: multiply top and bottom by the conjugate.
- **Uniqueness proofs**: take two objects with the property and combine them.

Traps

- The union of two subspaces is a subspace only when one contains the other (problem E1).
- $U_1 \cap (U_2 + U_3) \neq (U_1 \cap U_2) + (U_1 \cap U_3)$ in general (three lines in $\mathbb{R}^2$).
- Pairwise $U_i \cap U_j = \{0\}$ does not make $U_1 + U_2 + U_3$ direct.
- Injective and surjective are equivalent only for finite sets of equal size: $n \mapsto n+1$.
- Whether a set is a subspace depends on the field: $\mathbb{R}^2$ is closed under real multiples but not under multiplication by i (HW1 P6).
- $0 \cdot v = 0$ and $(-1)v = -v$ are theorems, not axioms.

§2 Problem set

HW1 P1. Let $f\colon X \to Y$. For $U \subseteq X$, $V \subseteq Y$ let $f(U) = \{f(x) : x \in U\}$ and $f^{-1}(V) = \{x \in X : f(x) \in V\}$.

(a) Prove $f(f^{-1}(V)) \subseteq V$.
(b) Prove $U \subseteq f^{-1}(f(U))$.
(c) Give examples where each inclusion is strict. Determine exactly when equality holds in (a) and in (b).

Hint: take an arbitrary element and unfold both definitions; for (c) think about surjective and injective.

HW1 P2 (LADR 1A.7). Show that $\dfrac{-1+\sqrt{3}\,i}{2}$ is a cube root of 1.

Hint: square it first, then multiply once more.

HW1 P3. Let $a, b \in \mathbb{F}$. Prove that if $ab = 0$ then $a = 0$ or $b = 0$.

Hint: multiplicative inverse (field axiom (vii)).

HW1 P4 (LADR 1B.2). If $a \in \mathbb{F}$, $v \in V$ and $av = 0$, prove $a = 0$ or $v = 0$.

Hint: same idea as P3, with Prop. 4(ii).

HW1 P5 (LADR 1C.4). Let $b \in \mathbb{R}$. Show that the set of continuous $f\colon [0,1] \to \mathbb{R}$ with $\int_0^1 f = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Hint: the zero function for one direction, linearity of the integral for the other.

HW1 P6 (LADR 1C.5). Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Hint: which scalars are allowed in $\mathbb{C}^2$?

HW1 P7. Over $\mathbb{F}_2$, expand $(x+1)^8$ by computing $(x+1)^2$ and squaring repeatedly. Explain where the calculation differs from the one over $\mathbb{R}$.

Hint: $2 = 0$ in $\mathbb{F}_2$.

R1 (Midterm 1 review, Exercise 7). Prove that the intersection of subspaces is a subspace. What about the union?

E1 (LADR 1C.12). Prove that the union of two subspaces of V is a subspace if and only if one of them contains the other.

E2. In $\mathbb{R}^3$ let $U = \{(x, y, 0)\}$, $W = \{(0, y, z)\}$, $X = \{(t, t, t)\}$. Which of the sums $U + W$, $U + X$, $W + X$ are direct? Is $\mathbb{R}^3 = U \oplus X$?